# Chemical Gardon

## Purpose

To observe the growth of the chemical garden and explain chemical reactions such as osmosis and liquid flow.

## Materials

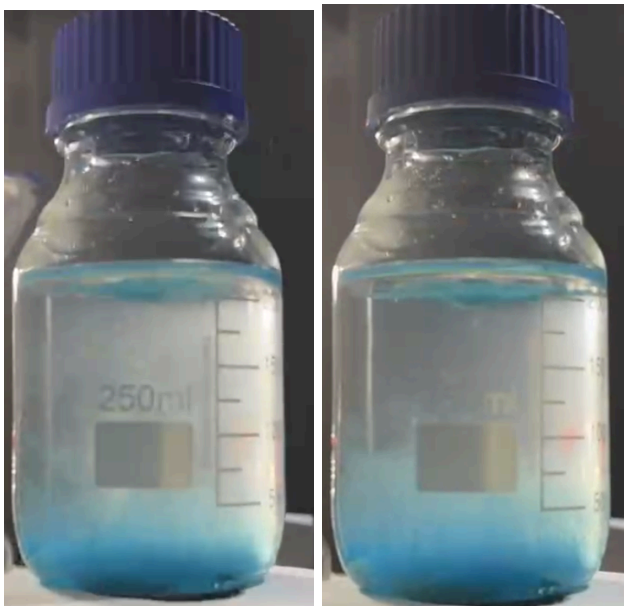
| Chemicals or equipment | Quantity |
|---|---|
| Copper(II) sulfate pentahydrate, $CuSO_4 \cdot 5H_2O$ | Small crystals |
| Sodium silicate solution | 40% stock solution; dilution and volume |
| Distilled water | |
| Clear beaker or test tube | 1 |
| Laboratory spoon | 1 |

## Procedure

1. Add 150 mL of distilled water
2. Add 50 mL of 40% sodium silicate solution, ratio 1: 3.
3. Wait for 5 minutes until there are no bubbles.
4. Add 20 g of copper sulfate powder on the surface.
5. Waite for 30minnutes
6. The recorded video shows the slow growth

## Observation

The blue “cloud” formed at the beginning, then dropped to the bottom. After 5 minutes, some tubes slowly grew from the crystal and bent with other branches at the middle parts,




## Chemistry explanation

- Cooper sulfate dissolved and reacted with the water glass. Formed a thin membrane layer.
- Water enters the membrane by osmosis, which increases the pressure inside.
- This pressure breaks the membrane, which allows the copper liquid to flow out of the membrane.
- $Cu^{2+} + 2OH^{-} \rightarrow Cu(OH)_2$
- The liquid flows out of the membrane and reacts with the solution to build the hollow tubes
- It is vulnerable to mixing or shifting of the glass because the flow of liquid makes the chemical reaction that bends the tubes together.